

Solutions to Axler's *Linear Algebra Done Right*, 4th Edition

by: blanketism

began: July 4, 2026

last edited: September 20, 2026

Contents

Preface	3
1 Vector Spaces	4
1A $\mathbb{R}^n$ and $\mathbb{C}^n$	4
1B Definition of Vector Space	9
1C Subspaces	14
2 Finite-Dimensional Vector Spaces	24
2A Span and Linear Independence	24
2B Bases	33
2C Dimension	40
3 Linear Maps	51
3A Vector Space of Linear Maps	51
3B Null Spaces and Ranges	62
3C Matrices	78
3D Invertibility and Isomorphisms	88
3E Products and Quotients of Vector Spaces	99
3F Duality	108
4 Polynomials	124
5 Eigenvalues and Eigenvectors	130
5A Invariant Subspaces	130
5B The Minimal Polynomial	146
5C Upper-Triangular Matrices	162
5D Diagonalizable Operators	169
5E Commuting Operators	182
6 Inner Product Spaces	187
6A Inner Products and Norms	187
6B Orthonormal Bases	205
6C Orthogonal Complements and Minimization Problems	220

7	Operators on Inner Product Spaces	231
7A	Self-Adjoint and Normal Operators	231
7B	Spectral Theorem	246
7C	Positive Operators	258
A	Appendix: Alternate Proofs of Exercises	269
A1	Vector Spaces	269
A2	Finite-Dimensional Vector Spaces	270
A3	Linear Maps	271
A4	Polynomials	276
A5	Eigenvalues and Eigenvectors	277
A6	Inner Product Spaces	280

7B Spectral Theorem

Exercise 7B1

Prove that a normal operator on a complex inner product space is self-adjoint if and only if all its eigenvalues are real.

Axler Note. This exercise strengthens the analogy (for normal operators) between self-adjoint operators and real numbers.

Solution.

$\Rightarrow$ This follows from 7.12.

$\Leftarrow$ Let T be a normal operator on a complex inner product space. By the complex spectral theorem, there exists an orthonormal basis such that $\mathcal{M}(T)$ is diagonal with respect to that basis. By the complex spectral theorem, the diagonal entries of $\mathcal{M}(T)$ are some $\lambda_1, \dots, \lambda_n$, and the diagonal entries of $\mathcal{M}(T^*)$ are $\overline{\lambda_1}, \dots, \overline{\lambda_n}$. If all the eigenvalues of T are real, then $\lambda_i = \overline{\lambda_i}$ for all i , so $\mathcal{M}(T) = \mathcal{M}(T^*)$, which implies $T = T^*$. ■

Exercise 7B2

Suppose $\mathbb{F} = \mathbb{C}$. Suppose $T \in \mathcal{L}(V)$ is normal and has only one eigenvalue. Prove that T is a scalar multiple of the identity operator.

Solution. By the complex spectral theorem, there exists an orthonormal basis such that $\mathcal{M}(T)$ is diagonal with respect to that basis. Since T has only one eigenvalue, all the diagonal entries of $\mathcal{M}(T)$ are equal to some λ . Therefore, $\mathcal{M}(T) = \lambda I$, which implies $T = \lambda I$. ■

Exercise 7B3

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$ is normal. Prove that the set of eigenvalues of T is contained in $\{0, 1\}$ if and only if there is a subspace U of V such that $T = P_U$.

Solution.

$\Rightarrow$ Suppose the set of eigenvalues of T is contained in $\{0, 1\}$. By the complex spectral theorem, there exists an orthonormal basis $e_1, \dots, e_n$ such that $\mathcal{M}(T)$ is diagonal with respect to that basis. Since all the eigenvalues are in $\{0, 1\}$, the diagonal entries of $\mathcal{M}(T)$ are all 0 or 1.

Let $U = \text{span}(\{e_i \mid \lambda_i = 1\})$. Then for any $v \in V$, we have

$$v = \sum_{j=1}^n \langle v, e_j \rangle e_j \implies Tv = \sum_{\lambda_j=1} \langle v, e_j \rangle e_j = P_U v,$$

where the last equality follows from the orthonormal basis.

$\Leftarrow$ If $T = P_U$, then by [Exercise 7A20](#), we know $P_U^2 = P_U$. If $Tv = \lambda v$ for some $\lambda \in \mathbb{C}$ and $v \neq 0$, then $\lambda^2 v = \lambda v$. This yields $\lambda^2 = \lambda$, so $\lambda \in \{0, 1\}$. ■

Exercise 7B4

Prove that a normal operator on a complex inner product space is skew (meaning it equals the negative of its adjoint) if and only if all its eigenvalues are purely imaginary (meaning that they have real part equal to 0).

Solution.

$\Rightarrow$ Let $v \in V$ be an eigenvector of T with eigenvalue λ . From 7.21(e), we know that $Tv = \lambda v$ implies $T^*v = \bar{\lambda}v$. If T is skew, then $T = -T^*$, so $Tv = -T^*v = -\bar{\lambda}v$. Therefore, $\lambda = -\bar{\lambda}$, which means λ is purely imaginary.

$\Leftarrow$ Suppose all eigenvalues of T are purely imaginary. By the complex spectral theorem, there exists an orthonormal basis $e_1, \dots, e_n$ such that $\mathcal{M}(T)$ is diagonal with respect to that basis. Let the diagonal entries of $\mathcal{M}(T)$ be $\lambda_1, \dots, \lambda_n$. Since each λ_i is purely imaginary, we have $\lambda_i = -\bar{\lambda}_i$. Therefore, the diagonal entries of $\mathcal{M}(T^*)$ are $\bar{\lambda}_1, \dots, \bar{\lambda}_n$, and the diagonal entries of $-\mathcal{M}(T)$ are $-\lambda_1, \dots, -\lambda_n$. Since $\bar{\lambda}_i = -\lambda_i$ for each i , we have $\mathcal{M}(T^*) = -\mathcal{M}(T)$, which implies $T^* = -T$. ■

Exercise 7B5

Prove or give a counterexample: If $T \in \mathcal{L}(\mathbb{C}^3)$ is a diagonalizable operator, then T is normal (with respect to the usual inner product).

Solution. This is false. Consider $T \in \mathcal{L}(\mathbb{C}^3)$ with matrix

$$\mathcal{M}(T) = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Since the diagonal entries are distinct, T is diagonalizable. However, $\|Te_1\| = 1$, while $\|T^*e_1\| = \sqrt{2}$, so T is not normal by 7.20. ■

Exercise 7B6

Suppose V is a complex inner product space and $T \in \mathcal{L}(V)$ is a normal operator such that $T^9 = T^8$. Prove that T is self-adjoint and $T^2 = T$.

Solution. By the complex spectral theorem, there exists an orthonormal basis $e_1, \dots, e_n$ of eigenvectors of T with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$. Define $Te_i = \lambda_i e_i$ for each $i = 1, \dots, n$. From $T^9 = T^8$, we have $\lambda_i^9 = \lambda_i^8$, which implies $\lambda_i^8(\lambda_i - 1) = 0$. This gives $\lambda_i \in \{0, 1\}$. By [Exercise 7B1](#), since all eigenvalues of T are real, T is self-adjoint. Moreover, since each eigenvalue λ_i satisfies $\lambda_i \in \{0, 1\}$, we have $\lambda_i^2 = \lambda_i$ for each i , so $T^2e_i = \lambda_i^2e_i = \lambda_i e_i = Te_i$ for each $i = 1, \dots, n$. Since T^2 and T agree on the basis, $T^2 = T$ by 3.4. ■

Exercise 7B7

Give an example of an operator T on a complex vector space such that $T^9 = T^8$ but $T^2 \neq T$.

Solution. Consider $T \in \mathcal{L}(\mathbb{C}^2)$ defined by $T(z_1, z_2) = (z_2, 0)$. Then $T^2 = 0$, so $T^9 = T^8 = 0$, but $T^2 = 0 \neq T$. ■

Exercise 7B8

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Prove that T is normal if and only if every eigenvector of T is also an eigenvector of T^* .

Solution.

⇒ This follows directly from 7.21(e).

⇐ If $V = \{0\}$, then T is trivially normal. Otherwise, assume $\dim V > 0$. By Schur's theorem, there exists an orthonormal basis $e_1, \dots, e_n$ of V with respect to which T has an upper-triangular matrix A . We show by induction on k that $A_{k,j} = 0$ for all $j \neq k$, so that A is diagonal. Suppose $A_{i,j} = 0$ for all $j \neq i$ and all $i < k$. Since A is upper-triangular, $Te_k = \sum_{i=1}^k A_{i,k}e_i$. By the inductive hypothesis, $A_{i,k} = 0$ for all $i < k$. Hence, $Te_k = A_{k,k}e_k$, so e_k is an eigenvector of T . By hypothesis, e_k is an eigenvector of T^* , say $T^*e_k = \lambda e_k$. Since $M(T^*) = A^*$, this gives

$$T^*e_k = \sum_{j=1}^n \overline{A_{k,j}}e_j = \lambda e_k,$$

and the uniqueness of the representation with respect to $e_1, \dots, e_n$ forces $\overline{A_{k,j}} = 0$ for all $j \neq k$. Thus, A is diagonal, so A and A^* commute, and therefore $TT^* = T^*T$. ■

Exercise 7B9

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Prove that T is normal if and only if there exists a polynomial $p \in \mathcal{P}(\mathbb{C})$ such that $T^* = p(T)$.

Solution.

⇒ If $V = \{0\}$, then $T^* = 0 = p(T)$ trivially with $p = 0$. Otherwise, suppose $V \neq \{0\}$. If T is normal, then by the complex spectral theorem, there exists an orthonormal basis of V consisting of eigenvectors of T . Let $e_1, \dots, e_n$ be such a basis, and let $\lambda_1, \dots, \lambda_n$ be the corresponding eigenvalues. Of the eigenvalues, let $\mu_1, \dots, \mu_m$ be the distinct ones. By Exercise 4-7, there exists a polynomial $p \in \mathcal{P}(\mathbb{C})$ such that $p(\mu_i) = \overline{\mu_i}$ for all $i \in \{1, \dots, m\}$. Since each eigenvalue λ_i is equal to some μ_j , we have $p(\lambda_i) = \overline{\lambda_i}$ for all i . Because T is normal and $Te_i = \lambda_i e_i$, we have $T^*e_i = \overline{\lambda_i} e_i$ by 7.21(e). Also, $p(T)e_i = p(\lambda_i)e_i = \overline{\lambda_i} e_i$. Hence, $T^*e_i = p(T)e_i$ for all i . Since T^* and $p(T)$ agree on a basis of V , we have $T^* = p(T)$ by 3.4.

⇐ Suppose $T^* = p(T)$ for some polynomial $p \in \mathcal{P}(\mathbb{C})$. Since T commutes with $p(T)$, we have $TT^* = Tp(T) = p(T)T = T^*T$, so T is normal. ■

Exercise 7B10

Suppose V is a complex inner product space. Prove that every normal operator on V has a square root.

Axler Note. An operator $S \in \mathcal{L}(V)$ is called a **square root** of $T \in \mathcal{L}(V)$ if $S^2 = T$. We will discuss more about square roots of operators in Sections 7C and 8C.

Solution. Let $T \in \mathcal{L}(V)$ be normal. By the complex spectral theorem, there exists an orthonormal basis $e_1, \dots, e_n$ of V consisting of eigenvectors of T with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$. For each i , the fundamental theorem of algebra applied to the polynomial $z^2 - \lambda_i$ asserts the existence of $\mu_i \in \mathbb{C}$ such that $\mu_i^2 = \lambda_i$. Let $S \in \mathcal{L}(V)$ be the operator defined by $Se_i = \mu_i e_i$ for all i . Then

$$S^2 e_i = S(\mu_i e_i) = \mu_i S e_i = \mu_i (\mu_i e_i) = \mu_i^2 e_i = \lambda_i e_i = T e_i$$

for all i . Since S^2 and T agree on a basis of V , we have $S^2 = T$. ■

Exercise 7B11

Prove that every self-adjoint operator on V has a cube root.

Axler Note. An operator $S \in \mathcal{L}(V)$ is called a **cube root** of $T \in \mathcal{L}(V)$ if $S^3 = T$.

Solution. Let $T \in \mathcal{L}(V)$ be self-adjoint. By 7.29 (for $\mathbb{F} = \mathbb{R}$) or 7.31 (for $\mathbb{F} = \mathbb{C}$ by normality of T), there exists an orthonormal basis $e_1, \dots, e_n$ of V consisting of eigenvectors of T with corresponding eigenvalues $\lambda_1, \dots, \lambda_n \in \mathbb{F}$. By 7.12, these eigenvalues $\lambda_1, \dots, \lambda_n$ are real. For each i , the polynomial $z^3 - \lambda_i$ has a real root μ_i . Let $S \in \mathcal{L}(V)$ be the operator defined by $Se_i = \mu_i e_i$ for all i . Then

$$S^3 e_i = S^2(\mu_i e_i) = S(\mu_i^2 e_i) = \mu_i^3 e_i = \lambda_i e_i = T e_i$$

for all i . Since S^3 and T agree on a basis of V , we have $S^3 = T$. ■

Exercise 7B12

Suppose V is a complex inner product space and $T \in \mathcal{L}(V)$ is normal. Prove that if S is an operator on V that commutes with T , then S commutes with T^* .

Axler Note. The result in this exercise is called **Fuglede's theorem**.

Manual Remark. The original exercise statement said "Suppose V is a complex vector space...". I believe this to be an oversight, since T^* is defined using the inner product.

Solution. By Exercise 7B9, we may write $T^* = p(T)$ for some polynomial p . Since S commutes with T , it also commutes with any polynomial in T , and hence with T^* . ■

Exercise 7B13

Without using the complex spectral theorem, use the version of Schur's theorem that applies to two commuting operators (take $\mathcal{E} = \{T, T^*\}$ in Exercise 6B20) to give a different proof that if $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$ is normal, then T has a diagonal matrix with respect to some orthonormal basis of V .

Solution. Normality of T implies that T and T^* commute. By Exercise 6B20, there exists an orthonormal basis of V with respect to which both T and T^* have upper-triangular matrices. By 7.9, $\mathcal{M}(T^*) = \mathcal{M}(T)^*$. Since $\mathcal{M}(T)^*$ is upper-triangular, $\mathcal{M}(T)$ must be lower-triangular. This forces $\mathcal{M}(T)$ to be diagonal. ■

Exercise 7B14

Suppose $\mathbb{F} = \mathbb{R}$ and $T \in \mathcal{L}(V)$. Prove that T is self-adjoint if and only if all pairs of eigenvectors corresponding to distinct eigenvalues of T are orthogonal and $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$, where $\lambda_1, \dots, \lambda_m$ denote the distinct eigenvalues of T .

Solution.

$\Rightarrow$ Since T is self-adjoint, it is normal. By 7.22, its eigenvectors corresponding to distinct eigenvalues are orthogonal. By the real spectral theorem, V has an orthonormal basis consisting of eigenvectors of T , so T is diagonalizable. By 5.55, this gives the desired decomposition of V .

$\Leftarrow$ In each $E(\lambda_i, T)$, apply the Gram–Schmidt process to obtain an orthonormal basis for that eigenspace. By hypothesis, these orthonormal bases for the eigenspaces corresponding to distinct eigenvalues are mutually orthogonal. The combined list of these orthonormal bases forms an orthonormal list of eigenvectors of T , hence is linearly independent by 6.25. Moreover, this list spans V because $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$. Thus, the list is an orthonormal basis of V consisting of eigenvectors of T , so T is self-adjoint by the real spectral theorem. ■

Exercise 7B15

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Prove that T is normal if and only if all pairs of eigenvectors corresponding to distinct eigenvalues of T are orthogonal and $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$, where $\lambda_1, \dots, \lambda_m$ denote the distinct eigenvalues of T .

Solution. The proof is similar to that of [Exercise 7B14](#), but using the complex spectral theorem instead of the real spectral theorem. ■

Exercise 7B16

Suppose $\mathbb{F} = \mathbb{C}$ and $\mathcal{E} \subseteq \mathcal{L}(V)$. Prove that there is an orthonormal basis of V with respect to which every element of $\mathcal{E}$ has a diagonal matrix if and only if S and T are commuting normal operators for all $S, T \in \mathcal{E}$.

Axler Note. This exercise extends the complex spectral theorem to the context of a collection of commuting normal operators.

Solution.

$\Rightarrow$ By assumption, there exists an orthonormal basis of V with respect to which every element of $\mathcal{E}$ has a diagonal matrix. For each $T \in \mathcal{E}$, the diagonal matrix $\mathcal{M}(T)$ commutes with its conjugate transpose $\mathcal{M}(T)^*$. By 7.9, we know $\mathcal{M}(T^*) = \mathcal{M}(T)^*$, so $\mathcal{M}(T)$ commutes with $\mathcal{M}(T^*)$, which implies that T is normal. By 3.43, for all $S, T \in \mathcal{E}$, we have

$$\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T) = \mathcal{M}(T)\mathcal{M}(S) = \mathcal{M}(TS),$$

so $ST = TS$.

$\Leftarrow$ Taking $S = T$ in the hypothesis gives that each $T \in \mathcal{E}$ is normal. By [Exercise 7B12](#) (Fuglede's theorem), for all $S, T \in \mathcal{E}$, $ST^* = T^*S$. Then

$$S^*T^* = (TS)^* = (ST)^* = T^*S^*,$$

so S^* and T^* commute. Letting $\mathcal{E}^* = \{T^* \mid T \in \mathcal{E}\}$, we see that $\mathcal{E} \cup \mathcal{E}^*$ is a set of commuting operators. By [Exercise 6B20](#), there exists an orthonormal basis of V with respect to which every element of $\mathcal{E} \cup \mathcal{E}^*$ has an upper-triangular matrix. For each $T \in \mathcal{E}$, both $M(T)$ and $M(T^*)$ are upper-triangular, and $M(T^*) = M(T)^*$ by 7.9. Hence, $M(T)$ is both upper-triangular and lower-triangular, which implies that $M(T)$ is diagonal. Therefore, there exists an orthonormal basis of V with respect to which every element of $\mathcal{E}$ has a diagonal matrix. ■

Exercise 7B17

Suppose $\mathbb{F} = \mathbb{R}$ and $\mathcal{E} \subseteq \mathcal{L}(V)$. Prove that there is an orthonormal basis of V with respect to which every element of $\mathcal{E}$ has a diagonal matrix if and only if S and T are commuting self-adjoint operators for all $S, T \in \mathcal{E}$.

Axler Note. This exercise extends the real spectral theorem to the context of a collection of commuting self-adjoint operators.

Solution.

⇒ Fix an orthonormal basis of V with respect to which every element of $\mathcal{E}$ has a diagonal matrix. For any $T \in \mathcal{E}$, the matrix of T with respect to this orthonormal basis is diagonal. Moreover, 7.9 shows that $M(T^*) = M(T)^* = M(T)$. Since the matrices are equal and over the same orthonormal basis, it follows that $T = T^*$, so T is self-adjoint. Additionally, for any $S, T \in \mathcal{E}$, the matrices of S and T with respect to this orthonormal basis are diagonal, and diagonal matrices commute. A similar argument from [Exercise 7B16](#) shows that $ST = TS$.

⇐ We proceed by strong induction on $\dim V = n$. If $n = 0$, the empty list is an orthonormal basis of V , and the statement holds trivially. Now assume that the result holds for any real inner product space of dimension less than n and every collection of commuting self-adjoint operators on it. If every $T \in \mathcal{E}$ is some scalar multiple of the identity, then any orthonormal basis of V will diagonalize every element of $\mathcal{E}$, and the statement holds. In particular, this holds even with $\mathcal{E} = \emptyset$.

Otherwise, choose $T \in \mathcal{E}$ that is not a scalar multiple of the identity. Let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of T . By the real spectral theorem and [Exercise 7B14](#), we may write

$$V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T),$$

where the eigenspaces are pairwise orthogonal. Note that if $m = 1$, then $V = E(\lambda_1, T)$ so that $T = \lambda_1 I$, contradicting our choice of T as not a scalar multiple of the identity. Therefore, $m \geq 2$. For any i , nontriviality of each eigenspace implies that $\dim E(\lambda_i, T) = n - \sum_{j \neq i} \dim E(\lambda_j, T) \leq n - 1$.

We now claim that each eigenspace $E(\lambda_i, T)$ is invariant under every $S \in \mathcal{E}$. To see this, let $v \in E(\lambda_i, T)$ and $S \in \mathcal{E}$. Then

$$T(Sv) = S(Tv) = S(\lambda_i v) = \lambda_i(Sv),$$

so $Sv \in E(\lambda_i, T)$. Hence, each eigenspace $E(\lambda_i, T)$ is invariant under every $S \in \mathcal{E}$.

Now consider each eigenspace under the inner product inherited from V , and let $\mathcal{E}_i = \{S|_{E(\lambda_i, T)} \mid S \in \mathcal{E}\}$. For any $u, w \in E(\lambda_i, T)$,

$$\langle S|_{E(\lambda_i, T)} u, w \rangle = \langle Su, w \rangle = \langle u, Sw \rangle = \langle u, S|_{E(\lambda_i, T)} w \rangle$$

so that each $S|_{E(\lambda_i, T)}$ is a self-adjoint operator on $E(\lambda_i, T)$. Furthermore, for any $S|_{E(\lambda_i, T)}, R|_{E(\lambda_i, T)} \in \mathcal{E}_i$, we have

$$S|_{E(\lambda_i, T)} R|_{E(\lambda_i, T)} = (SR)|_{E(\lambda_i, T)} = (RS)|_{E(\lambda_i, T)} = R|_{E(\lambda_i, T)} S|_{E(\lambda_i, T)},$$

so $\mathcal{E}_i$ is a commuting family of self-adjoint operators on $E(\lambda_i, T)$. Then the induction hypothesis applies to $E(\lambda_i, T)$ and $\mathcal{E}_i$, so there exists an orthonormal basis of $E(\lambda_i, T)$ with respect to which every element of $\mathcal{E}_i$ has a diagonal matrix. By [Exercise 7B14](#), concatenating these bases gives an orthonormal basis of V since the eigenspaces are pairwise orthogonal, and V is the direct sum of these eigenspaces. Now let $S \in \mathcal{E}$. For each basis vector e in this orthonormal basis, there exists an i such that e is one of the basis vectors of $E(\lambda_i, T)$. Since $\mathcal{M}(S|_{E(\lambda_i, T)})$ is diagonal with respect to the chosen orthonormal basis of $E(\lambda_i, T)$, it follows that $Se = \mu_e e$ for some $\mu_e \in \mathbb{R}$. Therefore, the matrix of S with respect to this orthonormal basis is diagonal. ■

Exercise 7B18

Give an example of a real inner product space V , an operator $T \in \mathcal{L}(V)$, and real numbers b, c with $b^2 < 4c$ such that

$$T^2 + bT + cI$$

is not invertible.

Axler Note. This exercise shows that the hypothesis that T is self-adjoint cannot be deleted in 7.26, even for real vector spaces.

Solution. Consider the real inner product space $V = \mathbb{R}^2$ and the operator $T \in \mathcal{L}(V)$ whose matrix with respect to the standard basis is

$$\mathcal{M}(T) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

so that $T^2 = -I$. Now choose $b = 0$ and $c = 1$. Then $b^2 = 0 < 4c = 4$, and

$$T^2 + bT + cI = T^2 + I = -I + I = 0,$$

which is not invertible. ■

Exercise 7B19

Suppose $T \in \mathcal{L}(V)$ is self-adjoint and U is a subspace of V that is invariant under T .

- Prove that $U^\perp$ is invariant under T .
- Prove that $T|_U \in \mathcal{L}(U)$ is self-adjoint.
- Prove that $T|_{U^\perp} \in \mathcal{L}(U^\perp)$ is self-adjoint.

Solution.

- For any $u \in U$ and $w \in U^\perp$, we have

$$\langle Tw, u \rangle = \langle w, Tu \rangle = 0$$

since $Tu \in U$. Hence, $Tw \in U^\perp$.

- For any $u_1, u_2 \in U$, we have

$$\langle T|_U u_1, u_2 \rangle = \langle Tu_1, u_2 \rangle = \langle u_1, Tu_2 \rangle = \langle u_1, T|_U u_2 \rangle.$$

Hence, $T|_U$ is self-adjoint.

- Applying the same argument as in part (b) to $U^\perp$ gives the conclusion. ■

Exercise 7B20

Suppose $T \in \mathcal{L}(V)$ is normal and U is a subspace of V that is invariant under T .

- (a) Prove that $U^\perp$ is invariant under T .
- (b) Prove that U is invariant under T^* .
- (c) Prove that $(T|_U)^* = (T^*)|_U$.
- (d) Prove that $T|_U \in \mathcal{L}(U)$ and $T|_{U^\perp} \in \mathcal{L}(U^\perp)$ are normal operators.

Axler Note. This exercise can be used to give yet another proof of the complex spectral theorem. (Use induction on $\dim V$ and the result that T has an eigenvector.)

Manual Remark. The ordering of the parts in this exercise is strange. The solution below will prove (b) first as (a) falls out naturally as a consequence.

Solution.

- (b) If $U = \{0\}$, the result is immediate since the zero subspace is invariant under any operator. Otherwise, let $e_1, \dots, e_n$ be an orthonormal basis of U . By definition of P_U , we have for all $v \in V$ that $P_U v = \sum_{i=1}^n \langle v, e_i \rangle e_i$. Moreover, the Pythagorean theorem gives $\|v\|^2 = \|P_U v\|^2 + \|v - P_U v\|^2$. Hence, $\|v\|^2 = \sum_{i=1}^n |\langle v, e_i \rangle|^2$ if and only if $v \in U$.

Now, for each $j = 1, \dots, n$, the invariance of U under T gives $Te_j \in U$. Hence,

$$\|Te_j\|^2 = \sum_{i=1}^n |\langle Te_j, e_i \rangle|^2.$$

By Bessel's inequality, we have for all $j = 1, \dots, n$ that

$$\|T^* e_j\|^2 \geq \sum_{i=1}^n |\langle T^* e_j, e_i \rangle|^2 = \sum_{i=1}^n |\langle e_j, Te_i \rangle|^2.$$

Since T is normal, we have $\|T^* e_j\| = \|Te_j\|$ for each $j = 1, \dots, n$. Summing over $j = 1, \dots, n$ gives

$$\sum_{j=1}^n \|Te_j\|^2 = \sum_{j=1}^n \|T^* e_j\|^2 \geq \sum_{j=1}^n \sum_{i=1}^n |\langle T^* e_j, e_i \rangle|^2 = \sum_{i=1}^n \sum_{j=1}^n |\langle e_j, Te_i \rangle|^2 = \sum_{i=1}^n \|Te_i\|^2,$$

where $Te_i \in U$ by the invariance of U under T . Since the first and last sums in the inequality are equal, equality must hold throughout. In particular,

$$\sum_{j=1}^n \left(\|T^* e_j\|^2 - \sum_{i=1}^n |\langle T^* e_j, e_i \rangle|^2 \right) = 0.$$

By Bessel's inequality, each term in the sum is nonnegative. Hence, each term must be zero, and

$$\|T^* e_j\|^2 - \sum_{i=1}^n |\langle T^* e_j, e_i \rangle|^2 = 0 \quad \text{for each } j = 1, \dots, n.$$

The above equality shows that $T^* e_j \in U$ for each $j = 1, \dots, n$. The linearity of T^* then implies that $T^* u \in U$ for all $u \in U$, so U is invariant under T^* .

- (a) Let $w \in U^\perp$ and $u \in U$. Using (b), $T^* u \in U$, so $\langle w, T^* u \rangle = \langle Tw, u \rangle = 0$. Hence, $Tw \in U^\perp$, and $U^\perp$ is invariant under T .

(c) By (b), we know $(T^*)|_U \in \mathcal{L}(U)$. Then for all $x, y \in U$, we have

$$\langle (T|_U)y, x \rangle = \langle Ty, x \rangle = \langle y, T^*x \rangle = \langle y, (T^*)|_U x \rangle.$$

Since the adjoint is unique, we conclude that $(T|_U)^* = (T^*)|_U$.

(d) For any $u' \in U$, (b) gives $T^*u' \in U$. Using (c), we have

$$(T|_U)(T|_U)^*u' = T(T^*u') = (TT^*)u' = (T^*T)u' = T^*(Tu') = (T|_U)^*(T|_U)u',$$

showing normality of $T|_U$.

Now, by (a), we have that $U^\perp$ is invariant under T . Then we may use the arguments in (b) and (c) with $U^\perp$ in place of U to conclude that $U^\perp$ is invariant under T^* . Additionally, the argument of the first paragraph with $U^\perp$ in place of U shows that $T|_{U^\perp}$ is normal. ■

Exercise 7B21

Suppose that T is a self-adjoint operator on a finite-dimensional inner product space and that 2 and 3 are the only eigenvalues of T . Prove that

$$T^2 - 5T + 6I = 0.$$

Solution. By either the real or complex spectral theorem, we may take an orthonormal basis of eigenvectors $e_1, \dots, e_n$ of T with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$. For each $i \in \{1, \dots, n\}$, we have $\lambda_i \in \{2, 3\}$. Then

$$(T^2 - 5T + 6I)e_i = (\lambda_i^2 - 5\lambda_i + 6)e_i = (\lambda_i - 2)(\lambda_i - 3)e_i = 0.$$

Since $T^2 - 5T + 6I$ evaluates to 0 on each basis vector e_i , the conclusion follows. ■

Exercise 7B22

Give an example of an operator $T \in \mathcal{L}(\mathbb{C}^3)$ such that 2 and 3 are the only eigenvalues of T and $T^2 - 5T + 6I \neq 0$.

Solution. Consider $T \in \mathcal{L}(\mathbb{C}^3)$ with matrix representation

$$\mathcal{M}(T) = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Since T is upper-triangular with diagonal entries 2, 2, 3, these are the eigenvalues of T . However,

$$(T - 2I)(T - 3I)e_2 = (T - 2I)(e_1 - e_2) = -e_1 \neq 0. \quad \blacksquare$$

Exercise 7B23

Suppose $T \in \mathcal{L}(V)$ is self-adjoint, $\lambda \in \mathbb{F}$, and $\varepsilon > 0$. Suppose there exists $v \in V$ such that $\|v\| = 1$ and

$$\|Tv - \lambda v\| < \varepsilon.$$

Prove that T has an eigenvalue λ' such that $|\lambda - \lambda'| < \varepsilon$.

Axler Note. This exercise shows that for a self-adjoint operator, a number that is close to satisfying an equation that would make it an eigenvalue is close to an eigenvalue.

Solution. If $\mathbb{F} = \mathbb{R}$, the real spectral theorem guarantees an orthonormal basis of eigenvectors. If $\mathbb{F} = \mathbb{C}$, then T is normal, since T is self-adjoint, and the complex spectral theorem guarantees an orthonormal basis of eigenvectors. In either case, we obtain an orthonormal basis of eigenvectors $e_1, \dots, e_n$ of T with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$. Hence, we may write $v = \sum_{i=1}^n \alpha_i e_i$ for scalars $\alpha_1, \dots, \alpha_n \in \mathbb{F}$. Since the basis is orthonormal and $\|v\| = 1$, we have $\sum_{i=1}^n |\alpha_i|^2 = 1$. Then

$$Tv - \lambda v = \sum_{i=1}^n \alpha_i \lambda_i e_i - \sum_{i=1}^n \alpha_i \lambda e_i = \sum_{i=1}^n \alpha_i (\lambda_i - \lambda) e_i.$$

By orthonormality of the basis, we have

$$\|Tv - \lambda v\|^2 = \sum_{i=1}^n |\alpha_i|^2 |\lambda - \lambda_i|^2.$$

Now, choose k such that

$$|\lambda - \lambda_k| = \min_{1 \leq i \leq n} |\lambda - \lambda_i|.$$

Let $\lambda' = \lambda_k$ so that λ' is an eigenvalue of T . Since $|\lambda - \lambda'| = |\lambda - \lambda_k| \leq |\lambda - \lambda_i|$ for all i , we have

$$|\lambda - \lambda'|^2 = |\lambda - \lambda_k|^2 \sum_{i=1}^n |\alpha_i|^2 \leq \sum_{i=1}^n |\alpha_i|^2 |\lambda - \lambda_i|^2 = \|Tv - \lambda v\|^2 < \varepsilon^2.$$

Taking square roots, we conclude that $|\lambda - \lambda'| < \varepsilon$. ■

Exercise 7B24

Suppose U is a finite-dimensional vector space and $T \in \mathcal{L}(U)$.

- (a) Suppose $\mathbb{F} = \mathbb{R}$. Prove that T is diagonalizable if and only if there is a basis of U such that the matrix of T with respect to this basis equals its transpose.
- (b) Suppose $\mathbb{F} = \mathbb{C}$. Prove that T is diagonalizable if and only if there is a basis of U such that the matrix of T with respect to this basis commutes with its conjugate transpose.

Axler Note. This exercise adds another equivalence to the list of conditions equivalent to diagonalizability in 5.55.

Solution. Since U is only assumed to be a finite-dimensional vector space, we require an inner product to discuss orthonormality in the $\Leftarrow$ direction of both parts. Hence, we prove the following lemma: there exists an inner product on U that makes any basis $e_1, \dots, e_n$ orthonormal.

Let $f_1, \dots, f_n$ denote the standard basis of $\mathbb{F}^n$, and equip $\mathbb{F}^n$ with the Euclidean inner product. Define $\varphi \in \mathcal{L}(U, \mathbb{F}^n)$ to be the unique linear map

$$\varphi(e_i) = f_i \quad \text{for } i = 1, \dots, n.$$

Then for any $u \in U$ and scalars $\alpha_1, \dots, \alpha_n \in \mathbb{F}$ such that $u = \sum_{i=1}^n \alpha_i e_i$, we have $\varphi(u) = \sum_{i=1}^n \alpha_i f_i$. Additionally, if $\varphi(u) = 0$, then $\alpha_i = 0$ for each i so that $u = 0$. Hence, φ is injective. As in [Exercise 6A2](#), applied verbatim to a linear map rather than a linear operator, we may define an inner product on U as

$$\langle u, v \rangle = \langle \varphi(u), \varphi(v) \rangle \quad \text{for all } u, v \in U.$$

Moreover, $\langle e_i, e_j \rangle = \langle f_i, f_j \rangle = \delta_{ij}$ for all i, j , so the basis $e_1, \dots, e_n$ is orthonormal with respect to this inner product. This completes the proof of the lemma.

- (a) $\Rightarrow$ Suppose T is diagonalizable. Then the matrix of T with respect to some basis $u_1, \dots, u_n$ consisting of eigenvectors of T is a diagonal matrix whose diagonal entries are eigenvalues of T . Since a diagonal matrix is equal to its transpose, the matrix of T with respect to the basis $u_1, \dots, u_n$ equals its transpose.

$\Leftarrow$ Suppose there is a basis $e'_1, \dots, e'_n$ of U such that the matrix of T with respect to $e'_1, \dots, e'_n$ equals its transpose. By the lemma, there exists an inner product on U that makes the basis $e'_1, \dots, e'_n$ orthonormal. Let $M(T)$ denote the matrix of T with respect to the basis $e'_1, \dots, e'_n$; similarly for $M(T^*)$. By 7.9, $M(T^*) = M(T)^*$, and since all entries of $M(T)$ are real, $M(T)^* = M(T)^t$. Moreover, the hypothesis gives $M(T)^t = M(T)$. Therefore, $M(T^*) = M(T)$ so that T is self-adjoint. By the real spectral theorem, T has a diagonal matrix with respect to some orthonormal basis, hence T is diagonalizable.

- (b) $\Rightarrow$ Suppose T is diagonalizable. Then there exists a basis $v_1, \dots, v_n$ of U consisting of eigenvectors of T with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$. By definition, $Tv_i = \lambda_i v_i$ for each $i = 1, \dots, n$. Let $A = M(T, (v_1, \dots, v_n))$ so that A is a diagonal matrix with diagonal entries $\lambda_1, \dots, \lambda_n$. It follows that A^* is also a diagonal matrix with diagonal entries $\overline{\lambda_1}, \dots, \overline{\lambda_n}$. Since diagonal matrices commute, we have $AA^* = A^*A$.

$\Leftarrow$ Suppose there is a basis $e''_1, \dots, e''_n$ of U such that the matrix of T with respect to $e''_1, \dots, e''_n$ commutes with its conjugate transpose. By the lemma, there exists an inner product on U that makes the basis $e''_1, \dots, e''_n$ orthonormal. For any $S \in \mathcal{L}(U)$, let $M(S)$ denote the matrix of S with respect to the basis $e''_1, \dots, e''_n$. By 7.9, $M(T^*) = M(T)^*$, and the hypothesis gives $M(T)M(T)^* = M(T)^*M(T)$. By 3.43, we have

$$M(TT^*) = M(T)M(T^*) = M(T)M(T)^* = M(T)^*M(T) = M(T^*)M(T) = M(T^*T),$$

so $TT^* = T^*T$, and hence T is normal. By the complex spectral theorem, T has a diagonal matrix with respect to some orthonormal basis of U , hence T is diagonalizable. ■

Exercise 7B25

Suppose that $T \in \mathcal{L}(V)$ and there is an orthonormal basis $e_1, \dots, e_n$ of V consisting of eigenvectors of T , with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$. Show that if $k \in \{1, \dots, n\}$, then the pseudoinverse $T^\dagger$ satisfies the equation

$$T^\dagger e_k = \begin{cases} \frac{1}{\lambda_k} e_k & \text{if } \lambda_k \neq 0, \\ 0 & \text{if } \lambda_k = 0. \end{cases}$$

Solution. Let $S = \{k \mid \lambda_k \neq 0\}$. For any $v \in V$, we can write

$$v = \sum_{i=1}^n \langle v, e_i \rangle e_i.$$

Then

$$Tv = \sum_{i=1}^n \langle v, e_i \rangle Te_i = \sum_{i=1}^n \langle v, e_i \rangle \lambda_i e_i.$$

It follows that $\text{null } T = \text{span}(e_j \mid j \notin S)$, and $\text{range } T = \text{span}(e_i \mid i \in S)$. If $k \notin S$, then e_k is orthogonal to $\text{range } T$, so $P_{\text{range } T} e_k = 0$, and hence $T^\dagger e_k = 0$. If $k \in S$, then $e_k \in \text{range } T$ so that $P_{\text{range } T} e_k = e_k$. Since $\langle e_k, e_j \rangle = 0$ for all $j \notin S$, and these vectors span $\text{null } T$, we have $\frac{1}{\lambda_k} e_k \in (\text{null } T)^\perp$. Since $T(\frac{1}{\lambda_k} e_k) = e_k$, we have that $(T|_{(\text{null } T)^\perp})^{-1} e_k = \frac{1}{\lambda_k} e_k$, and hence $T^\dagger e_k = \frac{1}{\lambda_k} e_k$. This is exactly the formula we wanted to show. ■